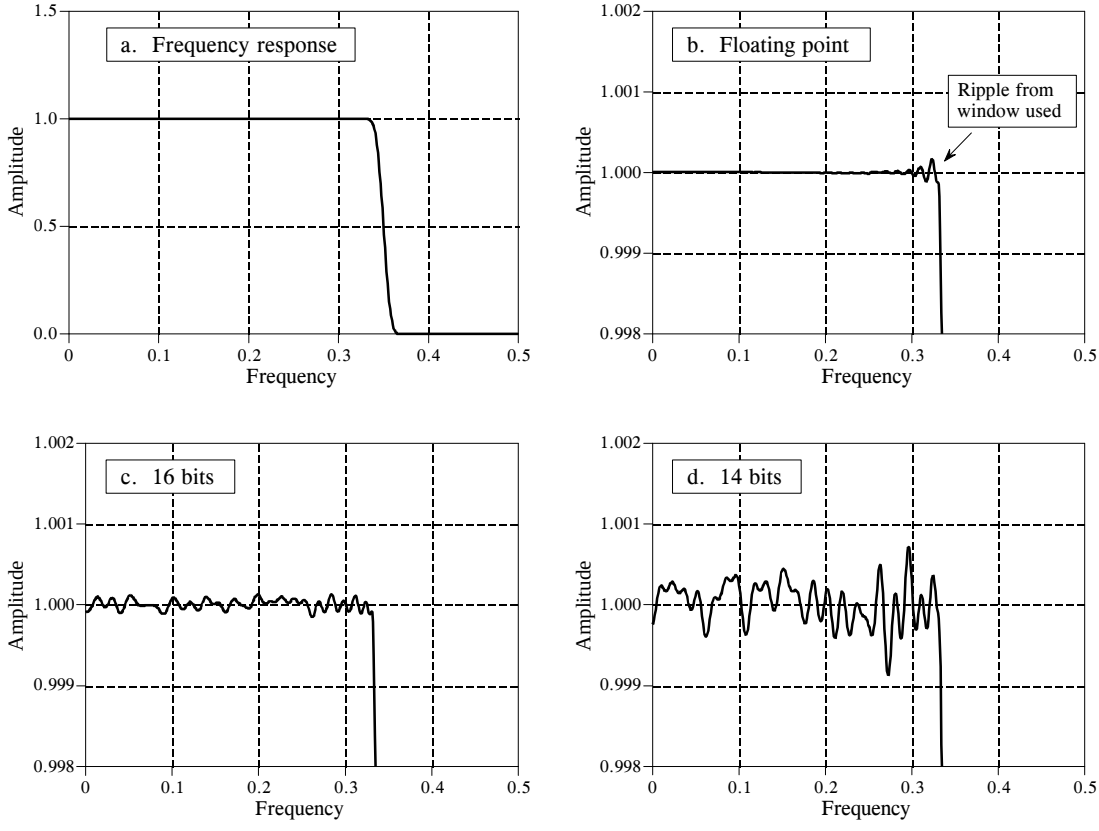

FIGURE 29-5
Round-off noise in the frequency response. Figure (a) shows the frequency response of a windowed-sinc low-pass filter, using a Blackman window and 150 points in the filter kernel. Figures (b), (c), and (d) show a more detailed view of this response by zooming in on the amplitude. When the filter kernel is represented in floating point, (b), the round-off noise is insignificant compared to the imperfections of the windowed-sinc design. As shown in (c) and (d), representing the filter kernel in fixed point makes round-off noise the dominate imperfection.

we must zoom in on the amplitude, as shown in (b), (c) and (d). As discussed in Chapter 3, this quantization appears much as additive random noise, limiting the detectability of small components in the signals.

These performance differences between fixed and floating point are often not important; for instance, they cannot even be seen in the "large scale" signals of Fig. 29-5a and Fig. 29-6a. However, there are some applications where the extra performance of floating point is helpful, and may even be critical. For instance, high-fidelity consumer audio system, such as CD players, represent the signals with 16 bit fixed point. In most cases, this exceeds the capability of human hearing. However, the best professional audio systems sample the signals with as high as 20 to 24 bits, leaving absolutely no room for artifacts that might contaminate the music. Floating point is nearly ideal for algorithms that process these high-precision digital signals.

Another case where the higher performance of floating point is needed is when the *algorithm* is especially sensitive to noise. For instance, FIR